

HAMILTONIAN ELEMENTS IN ALGEBRAIC K-THEORY

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ABSTRACT. A Hamiltonian fibration $M \hookrightarrow P \rightarrow X$ (with monotone compact fibers) induces a kind of “bundle of A_∞ categories” over X , with fiber given by the Fukaya category of M . Algebraic K -theory ideas, together with the above picture for $X = S^m$, yield the following: if R is a regular ring and G is a compact Lie group then there is a natural group homomorphism $\pi_m(BG) \rightarrow K_m(R) \oplus K_{m-1}(R)$. Given a Hamiltonian fibration as above, the construction generally first assigns elements in a certain categorified algebraic K -theory of R , analogous to Toën’s secondary K -theory. Taking the Langlands dual of G , one may ask for a relationship between the images of the homomorphisms above. This is thematically a K -theory expression of electric-magnetic duality.

1. INTRODUCTION

A guiding idea is that a Hamiltonian M -fibration over a smooth manifold X gives a kind of “bundle of A_∞ categories”, with fiber the Fukaya category of M . Exactly what this means is an interesting mathematical problem in its own right, and one answer is developed in [16]. Note that we are certainly not talking about local systems, as the “bundles” that one gets are generally non-trivial over spheres as shown in [15].

The second idea due to Toën [19], is that there is a refinement of K -theory of a commutative ring R , by replacing R -modules with dg or A_∞ categories over R , and using Waldhausen K -theory. We show that these two ideas naturally fit together. This culminates in a connection between symplectic geometry and algebraic K -theory, and in particular we get a certain geometric insight into what we call categorified algebraic K -theory of $\mathbb{Z}$. We are then naturally led to speculate on connections with Langlands duality phenomena.

Although in the main examples here the principal actors are compact groups and $PU(n)$ in particular, the framework is most naturally understood from the perspective of the group of Hamiltonian symplectomorphisms of some symplectic manifolds; so we quickly introduce this. Let (M, ω) be a symplectic manifold, throughout assumed to be compact, with ω a closed non-degenerate 2-form on M . The group of diffeomorphisms ϕ of M , satisfying $\phi^*\omega = \omega$, is called the symplectomorphism group of (M, ω) . When M is simply connected, this is also the group of Hamiltonian symplectomorphisms, which we denote by $\mathcal{H} = \text{Ham}(M, \omega)$. In general, a symplectomorphism is Hamiltonian if it is the time-one map of an isotopy generated by a family of vector fields $\{X_t\}$, $t \in [0, 1]$, satisfying

$$\omega(X_t, \cdot) = dH_t(\cdot),$$

for some family of smooth functions $\{H_t\}$ on M .

The group $\mathcal{H}$ is one of the more enigmatic objects of mathematics. It is always (unless $M = pt$) an infinite-dimensional Fréchet Lie group with its C^∞ topology. But it has a natural, bi-invariant Finsler metric called the Hofer metric. So in some ways it behaves like a compact group. Its topological and algebraic structure is tied to the main problems of Hamiltonian dynamics, see for instance Polterovich [11].

Much of the algebraic/topological structure of $\mathcal{H}$ is reflected in the properties of Hamiltonian fibrations, which are the main geometric ingredients in what follows. A *Hamiltonian fibration* $M \hookrightarrow P \rightarrow Y$ is

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a smooth fiber bundle over Y , with fiber a symplectic manifold (M, ω) , whose structure group is contained in $\text{Ham}(M, \omega)$. For example, take $(M, \omega) = (\mathbb{CP}^n, \omega_{FS})$ where ω_{FS} is the Fubini-Study symplectic form, and take the structure group $G = PU(n+1) \subset \text{Ham}(\mathbb{CP}^n, \omega_{FS})$.

The other ingredient is algebraic K -theory, which is a construction associating an infinite loop space ¹ $K(R)$ to any ring R , and so abelian groups:

$$K_m(R) = \pi_m(K(R)), \quad m \in \mathbb{N},$$

called K -theory groups of R . This is a powerful tool in algebraic topology and ring theory, with applications in number theory and geometric topology.

Taking R to be the non-commutative ring $\mathcal{K}$ of compact operators on a separable complex Hilbert space, the algebraic K -theory space $K(\mathcal{K})$ ² is identified with the topological K -theory space $KU = BU \times \mathbb{Z}$, see Karoubi [5], Suslin-Wodzicki [18]. In particular, a complex vector bundle E over a space X determines a homotopy class $[f_E : X \rightarrow K(\mathcal{K})]$, which we might call a *geometric element* of the corresponding abelian group. We shall see that in a certain sense this extends to a general ring R , provided it is commutative, and replacing a complex vector bundle by a suitable Hamiltonian fibration.

For each commutative ring R we construct “categorified algebraic K -theory groups” $K_m^{Cat}(R)$. Actually it will be convenient to define more general groups $K^{Cat}(X, R)$ for each simplicial set X , generated by aforementioned bundles of A_∞ categories over X with fibers A_∞ categories with finite type, see Definition 3.1. This is analogous to the group of classes $[Y \rightarrow K(\mathcal{K})]$ for Y a space, i.e. the topological K -theory of Y .

In principle, this categorified algebraic K -theory of R is a mostly minor variation of the secondary K -theory of R introduced by Toën, with the main difference that $\mathbb{Z}$ -graded chain complexes are replaced by $\mathbb{Z}_2$ -graded chain complexes. By way of the Hochschild chain functor, $K_m^{Cat}(R)$ should admit a natural map to the 2-periodic variant of algebraic K -theory of R , which we denote by $K_m^{\mathbb{Z}_2}(R)$, see Remark 3.7, although we don’t yet need this.

Remark 1.1. It was pointed out to me by Bertrand Toën that for R a regular ring, for example $R = \mathbb{Z}$, $K_m^{\mathbb{Z}_2}(R) \simeq K_m(R) \oplus K_{m-1}(R)$. This can be proved with A^1 homotopy theory.

To obtain “geometric elements”, we start with a Hamiltonian fibration $M \hookrightarrow P \rightarrow Y$, with fiber a monotone symplectic manifold. The following is a conjunction of Theorem 4.3 and Corollary 4.6.

Theorem 1.2. *If the Fukaya category of M is finite type, then for each R the data of the isomorphism class of P determines an element in $K^{Cat}(Y_\bullet, R)$. In particular, this gives mappings*

$$g_m : \pi_m(B \text{Ham}(\mathbb{CP}^n, \omega_{FS})) \rightarrow K_m^{Cat}(R).$$

Example 1. In the case of $n = 1$, in [15] using geometric arguments we outline the proof of non-triviality of g_4 for $R = \mathbb{Z}$. This stands in contrast to a theorem of Rognes [13] on the triviality of $K_4(\mathbb{Z})$.

Claim 1.3. $K_2^{Cat}(\mathbb{Z})$ is infinitely generated.

The proof of this is by using the mappings g_* to construct group injections

$$\mathbb{Z}_p \simeq \pi_2(BPU(p)) \rightarrow K_2^{Cat}(\mathbb{Z})$$

¹In this paper, the term infinite loop space is meant to be synonymous with connective Ω -spectrum, i.e. a sequence of spaces $\{E_n\}_{n \in \mathbb{N}}$ with $E_n \simeq \Omega E_{n+1}$. Then $K(R)$ is the notation for the 0-space E_0 of the corresponding spectrum.

²Rather one must take a certain non-connective variant of the algebraic K theory construction.

for all prime p . The details will appear elsewhere. Since the groups $K_m(\mathbb{Z})$ (and hence $K_m^{\mathbb{Z}_2}(\mathbb{Z})$ assuming Remark 1.1) are always finitely generated by foundational results of Quillen [12], the latter conclusion is perhaps surprising.

Our construction also gives the following, which in principle also lifts to categorified algebraic K -theory but this is blocked at the moment by open problems in the study of monotone Lagrangians of G/T .

Theorem 1.4. *For any compact Lie group G , there is a homotopy natural map:*

$$(1.5) \quad h : \mathrm{BG} \rightarrow K^{\mathbb{Z}_2}(R),$$

and so there are natural homomorphisms

$$h_* : \pi_m(\mathrm{BG}) \rightarrow K_m^{\mathbb{Z}_2}(R).$$

Taking the Langlands dual of G , we may ask about the correspondence of the respective mappings; we outline this in Section 6.

Question 1. Can one obtain the classes

$$[\mathrm{BG} \rightarrow K^{\mathbb{Z}_2}(R)]$$

by homotopy theory techniques, perhaps using tools of algebraic geometry? In other words, can we bypass the geometric analysis of the construction?

2. PRELIMINARIES

We will partly follow Keller [6], but our base commutative ring will be denoted R to avoid overloading notation. An A_∞ category C over R is a type of non-associative dg-category, with $\mathrm{hom}_C(a, b)$ a cochain complex over R . We refer the reader to standard references for further details, see for instance [9] which will come in use later.

Denote by $A_\infty \mathrm{Cat}_R^{\mathrm{strict}}$ the category of strictly unital, $\mathbb{Z}$ -graded A_∞ categories with morphisms strict A_∞ functors. We set $A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}}$ to be like $A_\infty \mathrm{Cat}_R^{\mathrm{strict}}$ but with chain complexes $\mathbb{Z}_n$ -graded instead of $\mathbb{Z}$ -graded. This means that all structure maps, functors, etc., are with respect to the $\mathbb{Z}_n$ grading. Various subscripts R may be omitted.

Let $\mathrm{Cat}_R^{\mathbb{Z}_n}$ denote the category of small R -linear $\mathbb{Z}_n$ -graded categories. For $C \in A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}}$, set $hC \in \mathrm{Cat}_R^{\mathbb{Z}_n}$ to be the category with the same objects and with $\mathrm{hom}_{hC}(a, b) = H_0(\mathrm{hom}_C(a, b))$.

Given a morphism $f : A \rightarrow B$ in $A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}}$, we denote by $f_* : hA \rightarrow hB$ the induced morphism in $\mathrm{Cat}_R^{\mathbb{Z}_n}$. We say that f is a *quasi equivalence* if f_* is an equivalence. We will say that f is a *Morita equivalence* if $T(f) : T(A) \rightarrow T(B)$ is a quasi equivalence, or if equivalently the induced functor $\mathcal{D}(B) \rightarrow \mathcal{D}(A)$ is an equivalence where $\mathcal{D}(A)$ denotes the derived category of A . We denote by $Hmo^{\mathbb{Z}_2}$ the homotopy category of $A_\infty \mathrm{Cat}_R^{\mathrm{strict}, \mathbb{Z}_2}$ with respect to Morita equivalences, this is a pointed category with the zero object: the category 0 with one object and zero endomorphism complex.

The category $A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}}$ is cocomplete, see [9]. Furthermore, it has a model category structure [10], which by left Bousfield localization yields a Morita model structure with weak equivalences Morita equivalences.

We will specifically need cofiber squares:

$$\begin{array}{ccc} A & \xrightarrow{I} & B \\ \downarrow & & \downarrow P \\ 0 & \longrightarrow & C, \end{array}$$

where I is a cofibration. Explicitly, a cofibration is a strict A_∞ functor, which is injective on objects and on hom sets is a cofibration of the corresponding complexes (in the projective model structure on complexes). Over a field this means that on hom sets the functor is a degree-wise split injection. A cofiber square as above represents the homotopy cofiber.

Definition 2.1. By a *short exact sequence* in $A_\infty \text{Cat}_R^{\text{strict}, \mathbb{Z}_2}$ we mean a sequence $A \xrightarrow{I} B \xrightarrow{P} C$ which fits into a cofiber square as above.

As explained by Keller, an explicit example for such short exact sequences come from Drinfeld dg-quotients.

We now briefly review simplicial sets. We denote by Δ the simplex category:

- The set of objects of Δ is $\mathbb{N}$.
- $\text{hom}_\Delta(n, m)$ is the set of non-decreasing maps $[n] \rightarrow [m]$, where $[n] = \{0, 1, \dots, n\}$, with its natural order.

A simplicial set X is a functor

$$X : \Delta^{op} \rightarrow \text{Set}.$$

$X(n)$ is called the set of n -simplices of X . Δ^d will denote a particular simplicial set: the standard representable d -simplex, with

$$\Delta^d(n) = \text{hom}_\Delta(n, d).$$

Definition 2.2. For X a simplicial set, $\Delta(X)$ will denote the *simplex category of X* . This is the category such that:

- The set of objects $\text{obj } \Delta(X)$ is the set of simplicial maps:

$$\Sigma : \Delta^d \rightarrow X, \quad d \geq 0.$$

- Morphisms $f : \Sigma_1 \rightarrow \Sigma_2$ are commutative diagrams in $s\text{-Set}$:

$$(2.3) \quad \begin{array}{ccc} \Delta^d & \xrightarrow{\quad} & \Delta^n \\ & \searrow \Sigma_1 & \downarrow \Sigma_2 \\ & & X. \end{array}$$

Given a morphism $f : X \rightarrow Y$ of simplicial sets, we set $\Delta f : \Delta(X) \rightarrow \Delta(Y)$ to be the functor defined on objects by $\Delta f(\Sigma) = f \circ \Sigma$. If $m : \Sigma_1 \rightarrow \Sigma_2$ is a morphism in $\Delta(X)$:

$$\begin{array}{ccc} \Delta^k & \xrightarrow{m} & \Delta^d \\ & \searrow \Sigma_1 & \downarrow \Sigma_2 \\ & & X, \end{array}$$

then obviously the diagram below also commutes:

$$\begin{array}{ccc} \Delta^k & \xrightarrow{m} & \Delta^d \\ & \searrow h_1 & \downarrow h_2 \\ & & Y, \end{array}$$

where $h_i = \Delta f(\Sigma_i) = f \circ \Sigma_i$, $i = 1, 2$. And so the latter diagram determines a morphism $\Delta f(m) : h_1 \rightarrow h_2$ in $\Delta(Y)$. Clearly, this determines a functor Δf as needed.

Definition 2.4. Let X be a Kan complex and C any category. Let $F_i : \Delta(X) \rightarrow C$, $i = 0, 1$ be functors. A **concordance** of F_0 to F_1 is a functor $\tilde{F} : \Delta(X \times I) \rightarrow C$ s.t. $\tilde{F}|_{\Delta(X \times \{0\})} = F_0$ and $\tilde{F}|_{\Delta(X \times \{1\})} = F_1$.

3. DEFINITION OF CATEGORIFIED ALGEBRAIC K-THEORY GROUPS OF R

We need to put some algebraic finiteness conditions on our A_∞ categories, otherwise any K -theory we construct will be trivial due to the Eilenberg swindle, the same reason that K -theory of infinite dimensional vector bundles is trivial. A natural requirement is that our A_∞ categories be smooth and proper or saturated in the sense of Kontsevich-Soibelman [8], as this is a dualizability condition, see [20]. In the $\mathbb{Z}_n$ -graded setting the analogue of this condition is more complex, we will instead work with a less restrictive notion, which is not too hard to check for geometric examples.

Set $\mathcal{A} = \mathcal{A}_R \subset A_\infty \text{Cat}_R^{\mathbb{Z}_2, \text{strict}}$ to be the full-subcategory whose objects are finite type, where the latter is defined as follows. And set $w\mathcal{A}_R \subset \mathcal{A}_R$ to be subcategory with the same objects and morphisms Morita equivalences.

Definition 3.1. We say that $B \in A_\infty \text{Cat}_R^{\mathbb{Z}_2, \text{strict}}$ is **finite type** if B is Morita equivalent to an A_∞ algebra, whose underlying chain complex in each degree is projective and finitely generated over R . Likewise, we ask that the Hochschild homology complex of B is projective and finitely generated over R .

Definition 3.2. Let X be a simplicial set. An A_∞ **bundle over** X is defined to be a functor $\Delta(X) \rightarrow w\mathcal{A}_R$.

Suppose we have A_∞ bundles over X , fitting into a sequence of natural transformations of functors to $\mathcal{A}_R$:

$$(3.3) \quad F_0 \rightarrow F_1 \rightarrow F_2,$$

s.t. for each object $\Sigma \in \Delta(S^k)$ the resulting sequence in $\mathcal{A}_R$ is a short exact sequence. Then we call $F_0 \rightarrow F_1 \rightarrow F_2$ a **short exact sequence**.

Definition 3.4. We define $K^{\text{Cat}}(X, R)$ to be the quotient of the free abelian group generated by the set of concordance classes $[F]$ of functors $F : \Delta(X) \rightarrow w\mathcal{A}_R$, by the subgroup generated by

$$\{[F_0] - [F_1] + [F_2] \mid \text{there is a short exact sequence } F_0 \rightarrow F_1 \rightarrow F_2\}.$$

This is clearly contravariantly functorial in X so if $f : X \rightarrow Y$ is a simplicial map then we get a group homomorphism $f^* : K^{\text{Cat}}(Y, R) \rightarrow K^{\text{Cat}}(X, R)$ defined on generators by $f^*([F]) = [F \circ \Delta f]$.

Remark 3.5. As classical algebraic K theory groups, this is covariantly functorial in R by “extension of scalars”.

Furthermore, it is a homotopy cofunctor in X , as by definition of concordance it is clear that if f is homotopic to g then $f^* = g^*$. In particular if X_0 is homotopy equivalent to X_1 then $K^{\text{Cat}}(X_0, R) = K^{\text{Cat}}(X_1, R)$.

Set $S_{\text{simp}}^k = \Delta^k / \partial \Delta^k$ for $\Delta^k / \partial \Delta^k$ the natural quotient Kan complex.

Definition 3.6. For $k > 0$, we define

$$K_k^{\text{Cat}}(R) = \ker\{(\Delta^0 \rightarrow S_{\text{simp}}^k)^* : K^{\text{Cat}}(S_{\text{simp}}^k, R) \rightarrow K^{\text{Cat}}(*, R)\},$$

and define

$$K_0^{\text{Cat}}(R) = K^{\text{Cat}}(\Delta^0, R).$$

Remark 3.7. The above definition of the groups $K_k^{Cat}(R)$ is meant to be the very direct, and amenable to computation. But from a theory perspective it would be great to explain why these groups are in fact the Waldhausen K -theory groups, that is homotopy groups of a spectrum associated to a certain Waldhausen category underlying $\mathcal{A}_R$. The needed category is already outlined in Toën in [19]. We take the full subcategory $\hat{\mathcal{A}}_R \subset \mathcal{A}_R$ with objects Morita fibrant (pretriangulated and idempotent complete). To be strict this will not be an Waldhausen category but ∞ -Waldhausen in the sense of Barwick [1], as for example the zero category 0 (the category with one object and zero complex as endomorphism set) is not initial in $\hat{\mathcal{A}}_R$ but the hom-sets $hom_{\hat{\mathcal{A}}_R}(0, A)$ are contractible when taken in the ∞ -localization of $\hat{\mathcal{A}}_R$ with respect to the Morita equivalences. The groups $K_k^{Cat}(R)$ are then exactly the groups $\pi_k(K(\hat{\mathcal{A}}_R))$ with $K(\hat{\mathcal{A}}_R)$ the Waldhausen K -theory spectrum. We postpone the details of this discussion to a future note.

3.1. A very basic computation. We recall the definition of $K_0(R)$. This is the quotient of the free abelian group generated by the set of isomorphism classes of projective finitely generated R -modules, by the subgroup generated by

$$\{[A] - [B] + [C] \mid \text{exists an exact sequence } 0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0.\}$$

If $\mathcal{E}$ is a triangulated category, one defines $K_0(\mathcal{E})$ to be the quotient of the free abelian group, generated by the set of isomorphism classes in $\mathcal{E}$, by the subgroup generated by

$$\{[A] - [B] + [C] \mid \text{exists a distinguished triangle } A \rightarrow B \rightarrow C \rightarrow A[1].\}$$

It is known that $K_k(D^{perf}(Mod_R)) \simeq K_k(R)$, where Mod_R denotes the category of projective finitely generated R -modules and $D^{perf}(Mod_R)$ denotes the derived category of perfect complexes.

If $D_n(Mod_R)$ denotes the n -periodic derived category, we no longer have that $K_0(R) \simeq K_0(D_n(Mod_R))$, as there are certain corrections, see Saito [14]. We do have:

Theorem 3.8 (Saito [14]). *For n even, the natural homomorphism:*

$$(3.9) \quad \psi : K_0(D_n(Mod_R)) \rightarrow K_0(R)$$

$$(3.10) \quad \psi([V]) = \sum_{i=0}^{i=n} (-1)^i [H_i(V)],$$

is an isomorphism.

There is a group homomorphism $K_0^{Cat}(\mathbb{Z}) \xrightarrow{\phi} K_0(D_2(Mod_{\mathbb{Z}}))$ induced by $[C] \mapsto [Hoch_{\bullet}(C)]$ for $Hoch_{\bullet}(C)$ the Hochschild homology complex of C . This readily follows by the fact that the Hochschild functor $Hoch_{\bullet}$ is “localizing” which means in this case that ϕ takes short exact sequence $C_0 \rightarrow C_1 \rightarrow C_2$ in $\mathcal{A}_R$ is taken to an exact triangle $Hoch_{\bullet}(C_0) \rightarrow Hoch_{\bullet}(C_1) \rightarrow Hoch_{\bullet}(C_2) \rightarrow Hoch_{\bullet}(C_0)[1]$, see Keller [6, Theorem 5.2].

Corollary 3.11. *The composition map*

$$K_0^{Cat}(\mathbb{Z}) \xrightarrow{\phi} K_0(D_2(Mod_{\mathbb{Z}})) \xrightarrow{\psi} K_0(\mathbb{Z}),$$

is non-trivial.

Proof. Take C to be a finite type $\mathbb{Z}_2$ -graded A_{∞} algebra over $\mathbb{Z}$ such that $HH_0(C) = \mathbb{Z}$, $HH_1(C) = 0$, $HH_2(C) = \mathbb{Z}$, where $HH_{\bullet}(C)$ is the homology of $Hoch_{\bullet}(C)$. As a geometric example take $C = FH_{\bullet}(L_0, L_0)$ where the latter is the Floer chain algebra of the equator L_0 in S^2 , then the Hochschild homology $HH_{\bullet}(C)$ is isomorphic to the singular homology $H_{\bullet}(S^2, \mathbb{Z})$, [17].

So we get

$$\psi([Hoch_{\bullet}(C)]) = 2[\mathbb{Z}] \neq 0 \in K_0(\mathbb{Z}) \simeq \mathbb{Z}.$$

□

A further corollary of the proof above is:

Corollary 3.12. $[\mathrm{Fuk}(S^2, \omega)]$ represents a non-trivial element in $K_0^{\mathrm{Cat}}(\mathbb{Z})$.

One natural question is how to produce interesting elements in $K_m^{\mathrm{Cat}}(R)$ for higher m . We will construct such elements using Hamiltonian fibrations and Floer theory.

4. A_∞ BUNDLES FROM HAMILTONIAN FIBRATIONS

The main construction of [16] has as input a smooth Hamiltonian fibration $M \hookrightarrow P \rightarrow Y$, with (M, ω) a compact monotone symplectic manifold, e.g. $(\mathbb{CP}^n, \omega_{FS})$. Denote by $Y_\bullet$ the simplicial set of smooth singular simplices $\Delta^n \rightarrow Y$ for Δ^n also denoting the standard topological n -simplex.

And the output is a functor

$$(4.1) \quad F_P : \Delta(Y_\bullet) \rightarrow wA_\infty \mathrm{Cat}_R^{\mathbb{Z}_2, \text{strict}},^3$$

whose concordance class F_P is independent of auxiliary choices.

Definition 4.2. We call a compact monotone symplectic manifold, whose monotone Fukaya category (see [17]) is finite type, *finite monotone type*.

To summarize, we obtain the following corollary of [16, Theorem 6.6].

Theorem 4.3. Let $M \hookrightarrow P \rightarrow Y$ be a Hamiltonian fibration with fiber (M, ω) finite monotone type. Then the assignment:

$$(4.4) \quad [P] \mapsto [F_P] \in K^{\mathrm{Cat}}(Y_\bullet, R)$$

is well defined, where $[P]$ is the Hamiltonian isomorphism class of the fibration P .

As an example, we have:

Proposition 4.5. $(\mathbb{CP}^n, \omega_{FS})$ is finite monotone type.

Proof. Specifically, we take a concrete model of $\mathrm{Fuk}(\mathbb{CP}^n, \omega_{FS})$ with objects oriented Lagrangians Hamiltonian isotopic to the Clifford torus L_0 . In this case it is automatic that $\mathrm{Fuk}(\mathbb{CP}^n, \omega_{FS})$ is quasi-isomorphic to the A_∞ algebra $HF(L_0, L_0)$, and finiteness follows. □

Example 2. More generally, given a compact Lie group G , G/T is always monotone with respect to some Kirillov-Kostant-Souriau symplectic structure ω_{KKS} . (The latter is well defined up to deformation equivalence, see for instance [7]). The study of some of its monotone Lagrangians is carried out for example in [3].

In particular, taking $Y = S^m$ and composing with the natural map

$$K^{\mathrm{Cat}}(S_\bullet^m, R) \simeq K^{\mathrm{Cat}}(S_{\mathrm{simp}}^m, R) \rightarrow K_m^{\mathrm{Cat}}(R)$$

we get:

Corollary 4.6. For each commutative ring R , there are natural maps:

$$(4.7) \quad g_m : \pi_m(B\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})) \rightarrow K_m^{\mathrm{Cat}}(R).$$

³In [16] we work with $R = \mathbb{Q}$ but this was only used for inverting quasi equivalences. As we don't need to do this here, the construction works over any commutative ring.

Proof. Let $\mathbb{CP}^n \hookrightarrow P \rightarrow B\text{Ham}(\mathbb{CP}^n, \omega_{FS})$ be the tautological Hamiltonian fibration, that is the fiber bundle associated to the universal principal $\text{Ham}(\mathbb{CP}^n, \omega_{FS})$ bundle over $B\text{Ham}(\mathbb{CP}^n, \omega_{FS})$. Use the conjunction of Proposition 4.5 and Theorem 4.3 to get the needed conclusion. $\square$

The map cl is a group homomorphism, but to show this we need to relate our definition of $K_k^{Cat}(R)$ to the Waldhausen construction, see Remark 3.7, and this will not be carried out here.

Remark 4.8. It is natural to expect that $Fuk(G/T, \omega_{KKS})$ is finite type. In this case we similarly get homomorphisms $\pi_m(\text{Ham}(G/T, \omega_{KKS})) \rightarrow K_m^{Cat}(R)$.

5. HAMILTONIAN ELEMENTS IN ALGEBRAIC K -THEORY

One of our goals is to obtain “Hamiltonian elements” of classical algebraic K -theory. Let $M \hookrightarrow P \rightarrow Y$ be a Hamiltonian fibration, with compact monotone fiber (M, ω) . Let

$$F_P : \Delta(X_\bullet) \rightarrow wA_\infty Cat_R^{\mathbb{Z}_2, strict}$$

be as in the previous section. Then the composition: $\Phi_P = Hoch_\bullet \circ F_P$ gives a functor to $wCh_R^{\mathbb{Z}_2}$, where $Ch_R^{\mathbb{Z}_2}$ is the Waldhausen category $Ch_R^{\mathbb{Z}_2}$ of 2-periodic complexes of projective finite rank R -modules. It’s Waldhausen K -theory infinite loop space will be denoted by $K^{\mathbb{Z}_2}(R)$.

For concreteness specialize to $Y = B\text{Ham}(G/T, \omega_{KKS})$, (from now on omitting ω_{KKS}) and P the associated G/T bundle over $B\text{Ham}(G/T)$. Note that by standard homotopy theory $|\Delta(Y_\bullet)| \simeq X$ where for a category C , $|C|$ is shorthand for the geometric realization of the nerve. Then we get a map:

$$|\Phi_P| : B\text{Ham}(G/T) \rightarrow |wCh_R^{\mathbb{Z}_2}|,$$

whose homotopy class is well defined by the fact that the concordance class of F_P is well defined.

Composing with the natural map $|wCh_R^{\mathbb{Z}_2}| \rightarrow K^{\mathbb{Z}_2}(R)$, coming from the construction of $K^{\mathbb{Z}_2}(R)$, see [21, Remark 8.3.2], we get a natural homotopy class:

$$(5.1) \quad [|\Phi_P| : B\text{Ham}(G/T) \rightarrow K^{\mathbb{Z}_2}(R)].$$

Proof of Theorem 1.4. Use the natural map

$$BG \rightarrow B\text{Ham}(G/T),$$

and (5.1). $\square$

6. LANGLANDS DUAL ALGEBRAIC K -THEORY ELEMENTS.

It is natural to ask how Langlands duality fits into the story of the mappings $\pi_m(BG) \rightarrow K_m(R) \oplus K_{m-1}(R)$. Denote by $K^\vee$ the maximal compact subgroup of the Langlands dual of the complexification $G_{\mathbb{C}}$. Via Theorem 1.4, we have natural maps:

$$\begin{aligned} \phi_G : \pi_m(BG) &\rightarrow K_m^{\mathbb{Z}_2}(R), \\ \phi_{K^\vee} : \pi_m(BK^\vee) &\rightarrow K_m^{\mathbb{Z}_2}(R). \end{aligned}$$

If G is simply laced it is not hard to see that the images of these maps are the same, as $G/T = K^\vee/T$ and the images of G and $K^\vee$ in $\text{Ham}(G/T)$ coincide. A tentative conjecture would be:

Conjecture 1. *For each compact G the images of ϕ_G and $\phi_{K^\vee}$ coincide.*

For $m = 0$ this is saying that $Fuk(G/T)$ and $Fuk(K^\vee/T)$ represent the same classes in $K_0^{\mathbb{Z}_2}(R)$. We might further ask if they represent the same classes in $K_0^{Cat}(R)$, which would be the case for instance if these categories were Morita equivalent (but is much weaker), and this is comparable to the homological mirror symmetry conjecture of Kontsevich.

Furthermore, there should be a lift, see Remark 4.8:

$$\begin{aligned}\tilde{\phi}_G : \pi_m(BG) &\rightarrow K_m^{Cat}(R), \\ \tilde{\phi}_{K^\vee} : \pi_m(BK^\vee) &\rightarrow K_m^{Cat}(R),\end{aligned}$$

and we can again ask if the images coincide.

If we take the general point of view of Kapustin-Witten [4], then the above conjecture is thematically a rudimentary K -theory expression of the electric-magnetic duality. One possibly related foundational work on topological aspects of Langlands duality is [2].

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